Literature Survey: Existing Research on Sign Language Recognition

Related work for the mini-project **SignVaani: Real-Time Indian Sign Language to Multilingual Speech**. Each entry lists what the paper does, the gap that SignVaani addresses, and clickable links (blue text) to the paper, dataset or code. Links were checked in October 2026.

A. Indian Sign Language (ISL) research

#	Paper, authors & venue	What it does	Gap SignVaani fills / use	Links
1	**INCLUDE: A Large Scale Dataset for Indian Sign Language Recognition** A. Sridhar, R. G. Ganesan, P. Kumar, M. M. Khapra *ACM Multimedia 2020 (DOI 10.1145/3394171.3413528)*	Word-level ISL dataset: 4,287 videos, 263 signs, 15 categories, signed by deaf students (Chennai). Also INCLUDE-50 subset; best reported 94.5% (INCLUDE-50) / 85.6% (full).	Lab dataset and recognition only; no live sentence building or speech output.	Paper (DOI) Dataset (Zenodo) Code
2	**OpenHands: Making Sign Language Recognition Accessible with Pose-based Pretrained Models across Languages** P. Selvaraj, G. NC, P. Kumar, M. M. Khapra *ACL 2022, pp. 2114-2133*	Uses pose keypoints as input; releases 4 pose-based models for 6 sign languages incl. ISL, and the largest pose-based ISL pretraining dataset.	Research toolkit, not an end-user real-time app; no Indian-language speech.	ACL Anthology arXiv
3	**CISLR: Corpus for Indian Sign Language Recognition** A. Joshi, A. Bhat, P. S, P. Gole, S. Gupta, S. Agarwal, A. Modi *EMNLP 2022, pp. 10357-10366*	Word-level ISL corpus with ~4,700-word vocabulary; one-shot baseline that transfers features learned from ASL.	One-shot recognition from video; no real-time spotting or speech pipeline.	ACL Anthology
4	**ISLTranslate: Dataset for Translating Indian Sign Language** A. Joshi, S. Agrawal, A. Modi *Findings of ACL 2023, pp. 10466-10475*	About 31k ISL-English sentence pairs from educational videos (ISLRTC etc.); pose-based transformer baseline (Pose-SLT).	Continuous translation accuracy still low; heavy models; no speech output.	arXiv Code / data
5	**iSign: A Benchmark for Indian Sign Language Processing** A. Joshi, R. Mohanty, M. Kanakanti, A. Mangla, S. Choudhary, M. Barbate, A. Modi *Findings of ACL 2024, pp. 10827-10844*	Large ISL-English benchmark (118k+ video-sentence pairs) with tasks such as SignVideo2Text, SignPose2Text, Text2Pose.	Benchmark, not a deployable assistant; no domain focus or offline speech.	arXiv Project page
6	**ISL-CSLTR: Indian Sign Language Dataset for Continuous Sign Language Translation and Recognition** R. Elakkiya, B. Natarajan *Mendeley Data, 2021 (DOI 10.17632/kcmpdxky7p.1)*	Sentence-level ISL corpus: 700 videos of 100 sentences by 7 signers, plus word-level images.	Small vocabulary and few signers; dataset only.	Mendeley Data
7	**Hierarchical Windowed Graph Attention Network and a Large Scale Dataset for Isolated Indian Sign Language Recognition** (see arXiv page for author list) *arXiv 2024*	Larger isolated ISL dataset (2,002 words, 40,033 videos) and a graph-attention model on skeleton data.	GPU-oriented model; no sentence generation or speech.	arXiv
8	**Nearest Neighbour Classification of Indian Sign Language Gestures using Kinect Camera** Z. A. Ansari, G. Harit *Sadhana (Springer), vol. 41, no. 2, pp. 161-182, 2016*	140 ISL signs from 18 subjects with a Kinect depth camera and k-NN; ~90.7% on fingerspelling subset.	Needs a special depth camera; static gestures; not real-time sentences.	IIT Jodhpur page

B. Sign language datasets & recognition (ASL and others)

#	Paper, authors & venue	What it does	Gap SignVaani fills / use	Links
9	**Word-level Deep Sign Language Recognition from Video: A New Large-scale Dataset and Methods Comparison (WLASL)** D. Li, C. Rodriguez, X. Yu, H. Li *WACV 2020*	WLASL: 2,000+ ASL words from 100+ signers; compares appearance (I3D) and pose-based (Pose-TGCN) models.	ASL only; models too heavy for laptop CPU real-time use.	arXiv Dataset
10	**MS-ASL: A Large-Scale Data Set and Benchmark for Understanding American Sign Language** H. R. Vaezi Joze, O. Koller *BMVC 2019*	25,000+ videos, 1,000 ASL signs, 200+ signers; signer-independent splits; I3D baseline.	ASL only; RGB video models need GPUs.	arXiv
11	**Continuous Sign Language Recognition: Towards Large Vocabulary Statistical Recognition Systems Handling Multiple Signers** O. Koller, J. Forster, H. Ney *Computer Vision and Image Understanding, vol. 141, pp. 108-125, 2015*	Statistical continuous SLR on RWTH-PHOENIX-Weather and SIGNUM; highlights multi-signer difficulty.	German Sign Language; pre-deep-learning pipeline.	RWTH-PHOENIX page
12	**Neural Sign Language Translation** N. C. Camgoz, S. Hadfield, O. Koller, H. Ney, R. Bowden *CVPR 2018*	Defines sign language translation as neural machine translation; releases RWTH-PHOENIX-Weather 2014T.	German Sign Language; offline research setting, no speech.	CVF Open Access (PDF) Code
13	**Sign Language Transformers: Joint End-to-end Sign Language Recognition and Translation** N. C. Camgoz, O. Koller, S. Hadfield, R. Bowden *CVPR 2020*	Transformer with CTC loss jointly learns recognition and translation; 21.80 BLEU-4 on PHOENIX14T.	Large model; not ISL; no real-time deployment.	arXiv Code
14	**SignBERT: Pre-Training of Hand-Model-Aware Representation for Sign Language Recognition** H. Hu, W. Zhao, W. Zhou, Y. Wang, H. Li *ICCV 2021*	Self-supervised pre-training on hand-pose tokens (masking and reconstruction), then fine-tuning; SOTA on MSASL, WLASL, etc.	Hands only; heavy pre-training; not ISL.	arXiv CVF Open Access

C. Pose / keypoint-based methods (same approach as SignVaani)

#	Paper, authors & venue	What it does	Gap SignVaani fills / use	Links
15	**Sign Pose-based Transformer for Word-level Sign Language Recognition (SPOTER)** M. Bohacek, M. Hruz *WACV Workshops 2022, pp. 182-191*	Lightweight transformer on 2D body landmarks with hand-local normalisation and joint-rotation augmentation; 63.18% on WLASL100.	ASL/LSA64; no ISL, no sentence or speech module.	CVF Open Access
16	**Skeleton Aware Multi-modal Sign Language Recognition (SAM-SLR)** S. Jiang, B. Sun, L. Wang, Y. Bai, K. Li, Y. Fu *CVPR Workshops 2021*	Skeleton graph network (SL-GCN) + SSTCN + RGB/depth; won 2021 signer-independent isolated SLR challenge (98.42% RGB).	Multi-modal ensemble is heavy; Turkish SL; not real-time on CPU.	arXiv Code

#	Paper, authors & venue	What it does	Gap SignVaani fills / use	Links
17	**Spatial Temporal Graph Convolutional Networks for Skeleton-Based Action Recognition (ST-GCN)** S. Yan, Y. Xiong, D. Lin *AAAI 2018*	Graph convolution over skeleton joints in space and time; foundation of many skeleton-based SLR models.	Action recognition, not sign language; possible future model for SignVaani.	arXiv

D. Survey papers (for the literature review chapter)

#	Paper, authors & venue	What it does	Gap SignVaani fills / use	Links
18	**Sign Language Recognition: A Deep Survey** R. Rastgoo, K. Kiani, S. Escalera *Expert Systems with Applications, vol. 164, 2021*	Reviews deep-learning SLR methods, modalities and datasets.	General survey; ISL coverage limited.	Google Scholar
19	**Quantitative Survey of the State of the Art in Sign Language Recognition** O. Koller *arXiv 2020*	Meta-study of ~300 SLR papers (1983-2020); notes lack of non-manual (face) features in large systems.	Motivates SignVaani's use of facial landmarks.	arXiv
20	**Sign Language Recognition, Generation, and Translation: An Interdisciplinary Perspective** D. Bragg et al. *ACM ASSETS 2019 (Best Paper)*	Interdisciplinary review (CV, NLP, HCI, linguistics, Deaf culture); calls for involving Deaf communities.	Motivates SignVaani's user study with ISL users.	arXiv

E. Tools & models used in SignVaani

#	Paper, authors & venue	What it does	Gap SignVaani fills / use	Links
21	**MediaPipe: A Framework for Building Perception Pipelines** C. Lugaresi et al. *arXiv 2019*	Google's framework for real-time perception pipelines (pose, hands, face).	Used for landmark extraction.	arXiv
22	**MediaPipe Hands: On-device Real-time Hand Tracking** F. Zhang et al. *arXiv 2020*	21-point 3D hand landmark model that runs in real time on mobile CPUs.	Provides the hand keypoints.	arXiv
23	**IndicTrans2: Towards High-Quality and Accessible Machine Translation Models for all 22 Scheduled Indian Languages** J. Gala et al. (AI4Bharat) *Transactions on Machine Learning Research (TMLR), 2023*	Open translation models for all 22 scheduled Indian languages; releases 230M-pair parallel corpus.	Used for English to Hindi/Telugu/Kannada translation.	arXiv
24	**Towards Building Text-to-Speech Systems for the Next Billion Users** G. K. Kumar, Praveen S V, P. Kumar, M. M. Khapra, K. Nandakumar *ICASSP 2023*	Open FastPitch + HiFi-GAN TTS models for 13 Indian languages.	Used for Indian-language speech output.	arXiv

Research gap summary

Existing ISL work provides valuable **datasets** (INCLUDE, CISLR, ISLTranslate, iSign, ISL-CSLTR) and **research models** (OpenHands, graph networks), while the strongest pose-based methods (SPOTER, SAM-SLR, SignBERT) target ASL or other sign languages. None of the reviewed works combines: (1) real-time ISL recognition on a laptop CPU, (2) hand + body + face landmarks, (3) automatic sign spotting with an idle class, (4) ISL-grammar-aware sentence generation, (5) speech output in Indian languages, (6) offline, privacy-preserving operation, and (7) signer-independent evaluation with real ISL users. **SignVaani is designed to fill this combined gap.**

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